

Experiment 'p_3_5_order' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_5_order

Natural language statement: null

Method signature: p_3_5_order (i: int, j: int, k: int) returns (s: string)

Ensures

- if $i < j < k$ then $s == \text{"increasing"}$ else if $i > j \ \&\& \ j > k$ then $s == \text{"decreasing"}$ else $s == \text{"neither"}$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_5_order (i: int, j: int, k: int) returns (s: string)`

The method should respect the following contract:

`ensures if i < j < k then s == "increasing" else if i > j && j > k then s == "decreasing"`
 ↪ `else s == "neither"`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3A9vhWRJPb4tpch1JghSzD7mhgd

Model: gpt-5.5-2026-04-23

Created at: 1780582601

Finish reason: stop

Usage: 320 tokens in, and 193 tokens out

```
{
  if i < j && j < k {
    s := "increasing";
  } else if i > j && j > k {
    s := "decreasing";
  } else {
    s := "neither";
  }
}
```

Final Program

```
method p_3_5_order (i: int, j: int, k: int) returns (s: string)
  ensures if i < j < k then s == "increasing" else if i > j && j > k then s == "
    ↪ decreasing" else s == "neither"
{
  if i < j && j < k {
    s := "increasing";
```

```
} else if i > j && j > k {  
    s := "decreasing";  
} else {  
    s := "neither";  
}  
}
```

Total Token Usage

Input tokens: 320

Output tokens: 193

Reasoning tokens: 114

Sum of 'total tokens': 513

Experiment Timings

Overall Experiment started at 1780582600827, ended at 1780582671567, lasting 70740ms (70.74 seconds)

Iteration #1 started at 1780582600830, ended at 1780582671567, lasting 70737ms (70.74 seconds)